

Muhammad Usama

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INTRODUCTION

AI Researcher in 3D computer vision, CAD reverse-engineering, and generative CAD. 4+ years of industry and research experience taking ML systems from research to production, with publications at A* conferences.

EXPERIENCE

- **Toyota Research Institute (TRI)** Los Altos, United States
Intern Research Scientist Jun 2026 – Present
 - **Generative Reverse Engineering:** Architecting a novel vision-based framework for reverse-engineering automotive parts into highly editable CAD models that preserve design intent, with the goal of submission to CVPR 2027.
 - **Parametric CAD Generation:** Developing a NURBS-based generation framework for Toyota Motor Corporation (TMC) to translate non-editable geometry (“dumb solids”) into fully parametric, boundary representation (BRep) CAD models for automotive car roofs.
 - **Deformed CAD Pipeline (POC):** Engineered an end-to-end Python pipeline using OpenCASCADE and Trimesh to bridge the gap between simulation-based mesh morphing and CAD updates. Implemented algorithms to map tessellated mesh triangles to BRep face IDs, compute displacements, and apply global affine transformations.
- **German Research Center for Artificial Intelligence (DFKI)** Kaiserslautern, Germany
Research Assistant Nov 2023 – May 2026
 - **Automated captioning pipeline:** Built an end-to-end pipeline to auto-caption **~8.9M 3D assets** using vision-language models (VLMs); orchestrated multi-view renders and metadata extraction with Blender.
 - **Fine-tuning LLMs:** Researched and implemented **fine-tuning** strategies for large language models (LLMs), including Qwen-3B and Qwen-7B, using parameter-efficient methods such as **LoRA**.
 - **Long Context Fine-tuning:** Explored techniques for long-context fine-tuning of LLMs, evaluating approaches like **LongLoRA** and other context-extension strategies to enhance model performance on long-sequence tasks.
 - **Ethical filtering:** Designed an LLM-based ethical filtering layer with prompt engineering to detect, redact, and rewrite problematic captions; reduced ethical concerns by **~80%** relative to the unfiltered output.
 - **Visual Machine Learning:** Built backend **microservices** for a visual ML platform to create, train, and deploy models, supporting both the DFKI cluster and on-prem MindGarage CPU nodes.
- **Retrocausal** Redmond, United States
Machine Learning / Infrastructure Engineer Jul 2022 – Oct 2023
 - **Software Packaging:** Led the software packaging for client deployment, tailoring systems to diverse factory environments. This played a key role in expanding our client base and securing **venture capital funding in late 2023**, marking a significant career milestone that reflects the global scale and recognition of my work.
 - **System Scalability:** Improved system scalability and efficiency by replacing continuous polling with **Redis-** and **SQS-**based messaging, enabling on-demand data loading and significantly reducing API calls for long-running and continuous tasks.
 - **Labelling Tool:** Built an efficient object labelling method for the system, enabling accurate labelling of 2D static and 3D objects in depth and RGB frames. Designed an intuitive UI, resulting in time savings and enhanced product appeal.
 - **On-Premises Architecture Design and Development:** At a major client’s request, migrated the architecture from cloud to **on-premises** with minimal performance loss, replacing AWS with **LocalStack** and alternatives to Firebase and other cloud services, and built a packaging script for seamless installation on client PCs.
 - **Thin Web Client:** Developed a lightweight web client version to replace the bulky desktop application UI. Implemented **WebRTC** for seamless streaming and data channel communication.

PUBLICATIONS

- **NURBGen: Muhammad Usama**, Mohammad Sadil Khan, Didier Stricker, and Muhammad Zeshan Afzal (2026). *NURBGen: High-Fidelity Text-to-CAD Generation through LLM-Driven NURBS Modeling*. [AAAI Conference on Artificial Intelligence, AAAI 2026](#)
- **BRepCLIP: Muhammad Usama**, Didier Stricker, Mohammad Sadil Khan, and Muhammad Zeshan Afzal (2026). *BRepCLIP: Contrastive Multimodal Pretraining on BRep Primitives for CAD Understanding*. [Under Review, 2026](#)
- **MARVEL-40M+: Sankalp Sinha**, Mohammad Sadil Khan, **Muhammad Usama**, Shino Sam, Didier Stricker, Sk Aziz Ali, and Muhammad Zeshan Afzal (2025). *MARVEL-40M+: Multi-Level Visual Elaboration for High-Fidelity Text-to-3D Content Creation*. [Computer Vision and Pattern Recognition, CVPR 2025](#)
- **DreamCAD: Mohammad Sadil Khan**, **Muhammad Usama**, Rolandos Alexandros Potamias, Didier Stricker, Muhammad Zeshan Afzal, Jiankang Deng, and Ismail Elezi (2026). *DreamCAD: Scaling Multi-modal CAD Generation using Differentiable Parametric Surfaces*. [European Conference on Computer Vision, ECCV 2026](#)
- **Physics-in-the-Loop: Elias Berger**, **Muhammad Usama**, Jan Mehlstaubl, Bernhard Saske, and Kristin Paetzold-Byhain (2026). *Physics-in-the-Loop: A Hybrid Agentic Architecture for Validated CAD Engineering Design*. [International Joint Conference on Artificial Intelligence, IJCAI 2026](#)

EDUCATION

- **Technische Universität Kaiserslautern-Landau (RPTU)** Kaiserslautern, Germany
Master of Science in Computer Science Oct 2023 – Oct 2026
- **National University of Computer and Emerging Sciences (NUCES)** Karachi, Pakistan
Bachelor of Science in Computer Science Aug 2018 – Jul 2022

PROJECTS

- **Human Image Generation through Multimodal Diffusion Models** Jan 2024 – Jul 2024
Research Project
 - Seminar – Literature Review: To ground the project, surveyed 5 recent diffusion methods for pose-guided human image generation and style transfer (PIDM, PoCoLD, UPGPT, DiffBody, PCDMs) and benchmarked them against 3 GAN baselines on DeepFashion and Market-1501 (FID, SSIM, LPIPS), showing diffusion models lead on image quality.
 - HOI Dataset: Existing human-object interaction (HOI) datasets lacked object annotations, non-interaction states, and detailed captions. Built a multi-stage pipeline (SAM2, SDXL inpainting, DensePose, InternVL2-40B + LLaMA-70B) producing a **29K-image HOI dataset**, and curated **3,236** high-quality samples from **8,261** human-scored images.
 - HOIGEN Model: To generate interactions that preserve an object's exact appearance, proposed **HOIGEN**, a latent-diffusion inpainting model with a dedicated object feature encoder injected through attention; trained on the curated data, it generates structurally coherent, style-controllable, photorealistic HOI images from text and object appearance.

SKILLS

- **Programming Languages:** Python, C/C++, Java, JavaScript, C#, Dart
- **ML Architecture:** VLM-LLM Inference, Transformers, Diffusion Model, Stable Diffusion, LoRA
- **Software Engineering Frameworks:** .NET Core, Django, Spring Boot, ExpressJS, NestJS, Flutter, Java Android Development
- **ML Libraries:** PyTorch, PyTorch Lightning, PythonOCC, vLLM, Blender Python
- **Interpersonal:** Creative Thinker, Problem solver, Collaborative Engineer and Researcher, Quick and Adaptive Learner

HOBBIES

- **Sports:** Playing table tennis and cricket.
- **Travelling and learning:** Visited multiple cities across the United States, Greece, Saudi Arabia, Germany, France, Luxembourg, Singapore, and Pakistan, exploring local culture and history.